

Branch handling: the core algorithmic difference

LIVER

*one compact object, tracked as-is
(no branches to resolve)*



point on 1 slice



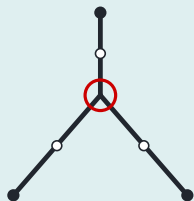
image-mode mask



propagate through z

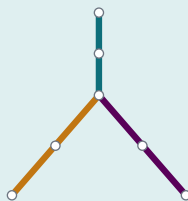
WORM

a neuron branches, so it is cut into single-object chains before propagating



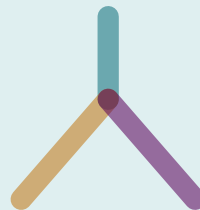
branching skeleton
○ branch point

DFS split



3 maximal linear chains
(+ virtual nodes)

propagate
each,
then merge



recomposed
per-neuron volume

Why the extra machinery: SAM2's video memory keeps following one arm at a branch, so a whole branching neuron cannot be tracked as a single object. Cutting it into linear chains first sidesteps this. The liver pipeline never needs this step because its structures are single compact objects.